

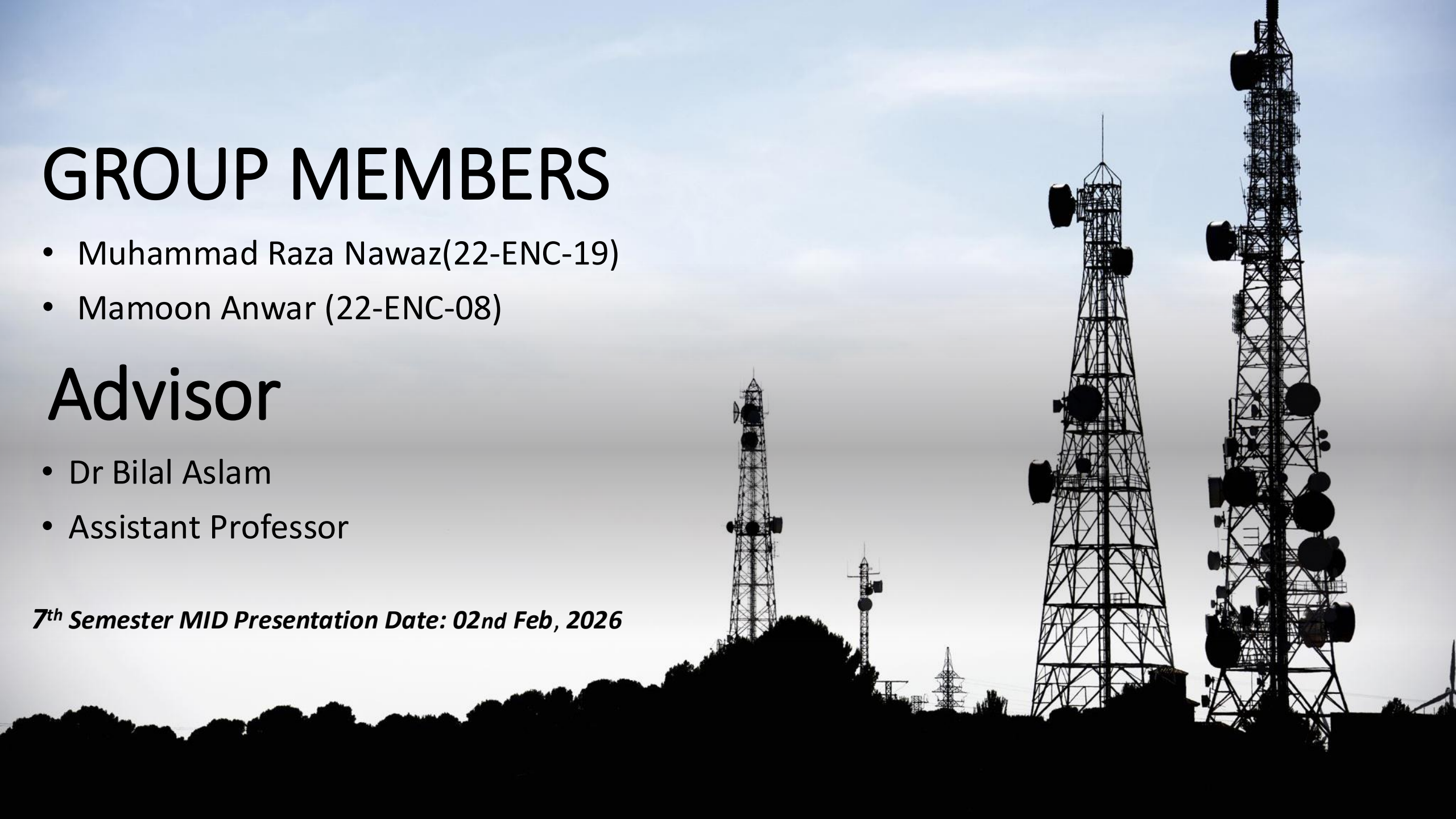
GROUP MEMBERS

- Muhammad Raza Nawaz(22-ENC-19)
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Advisor

- Dr Bilal Aslam
- Assistant Professor

7th Semester MID Presentation Date: 02nd Feb, 2026



Compact-planar Circularly Polarized antenna for emerging Internet of Things applications

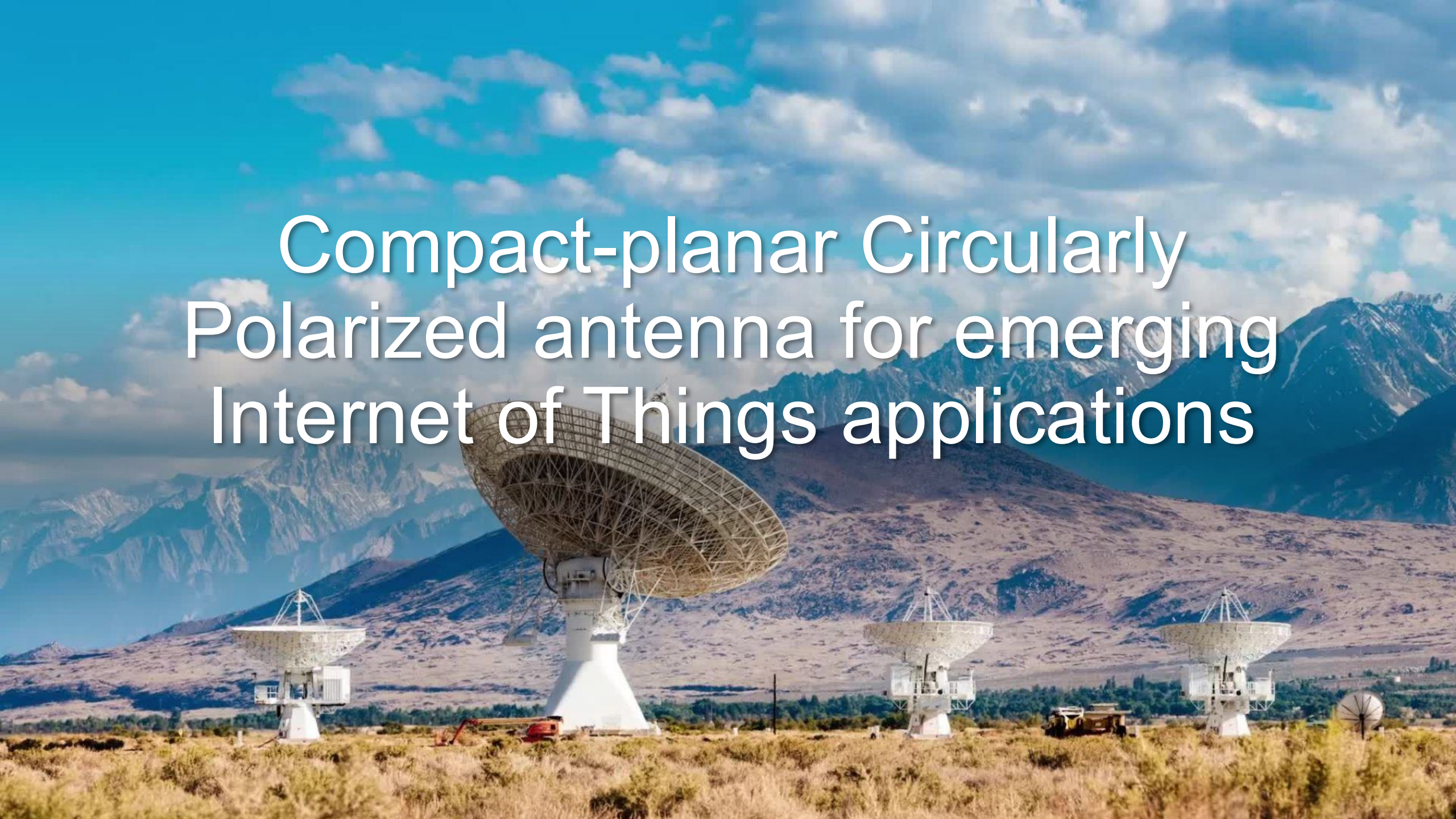


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Motivation

- Rapid development of **5G** significantly enhances wireless communication performance
- **Higher data rate, improved connectivity, and better spectral efficiency** compared to previous generations
- **Reduced latency**, enabling real-time and mission-critical applications
- **5G data rate is nearly 10× higher than 4G**, supporting high-speed communication
- Massive device connectivity enables **reliable Internet of Things (IoT)** applications
- mmWave bands suffer from **high path loss and atmospheric absorption**. These challenges create a strong need for **high-gain and wideband antenna designs**



Aims

- The primary aim of this project is to design, validate, and realize a highly integrable, compact, and planar circularly polarized antenna to meet the requirements of emerging wideband Internet of Things applications in the millimeter-wave band.

Objectives

- Design a **simple planar-structured antenna** with single-series-fed linear array geometry
- Enable **easy fabrication and system integration**
- Implement **simple circular polarization (CP) excitation** using geometric techniques
- Obtain **wide impedance bandwidth (~8 GHz)**
- Maintain **stable axial ratio below 3 dB**
- Achieve **stable, high broadside gain** across the operating band.



Introduction

The rapid growth of 5G and IoT requires high data rates and reliable wireless communication. To meet these demands, 5G operates in 2 bands millimeter-wave band (25–70 GHz) & sub-6Ghz, millimeter-wave suffers from high path loss, attenuation, and polarization mismatch. This project focuses on designing a compact, planar circularly polarized antenna with high gain and wide bandwidth to ensure stable and reliable 5G IoT communication in sub 6Ghz which provide a balance of both data rate & range.

Research Gap

No prior work sufficiently balances **planarity, simplicity of feed, wide impedance bandwidth, and wide AR bandwidth** for a fully integrable mm-Wave IoT device.



Methodology

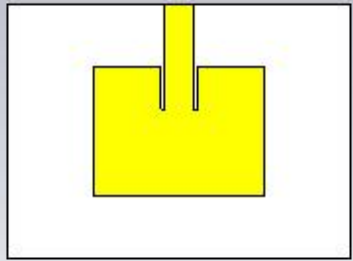


Phase 1: Preparation and Foundational Skills (✓)

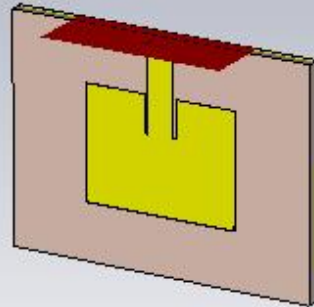
- Software Familiarization (CST/HFSS): Hands-on EM simulation practice ✓
- Basic Antenna Modeling: Design & simulation of patch and dipole antennas ✓
- Literature Review: Detailed study of the reference CP antenna paper ✓

CST Design

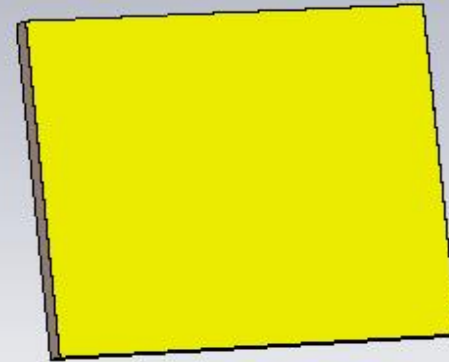
Front Side



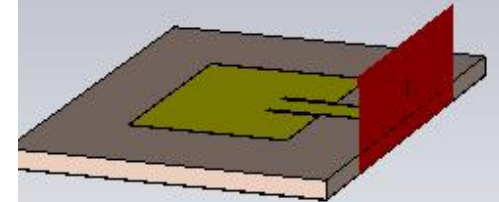
3d Perspective



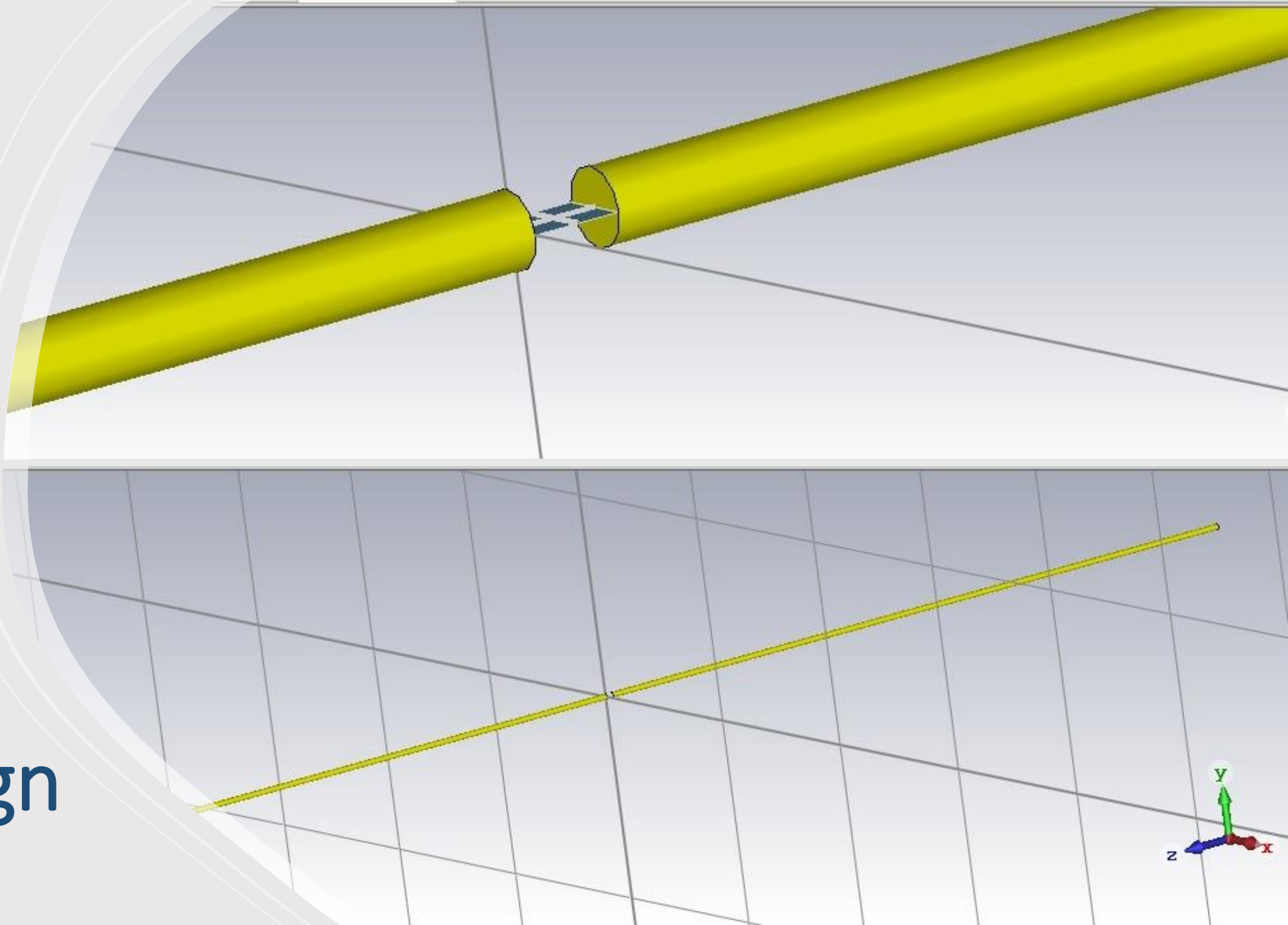
Back Side



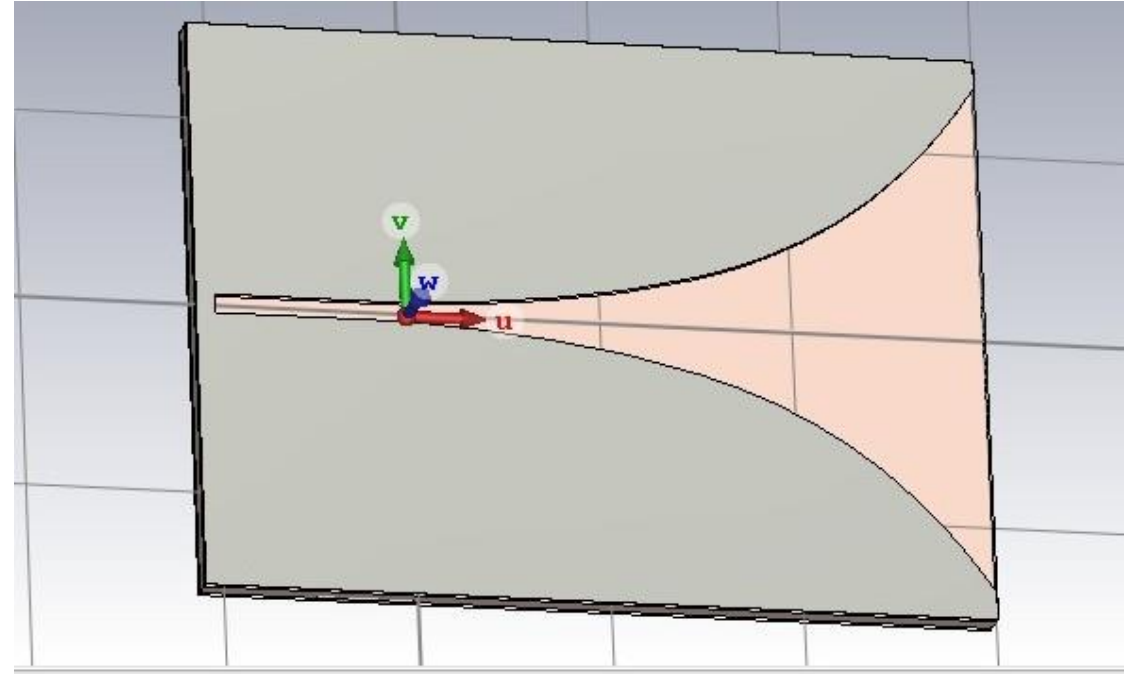
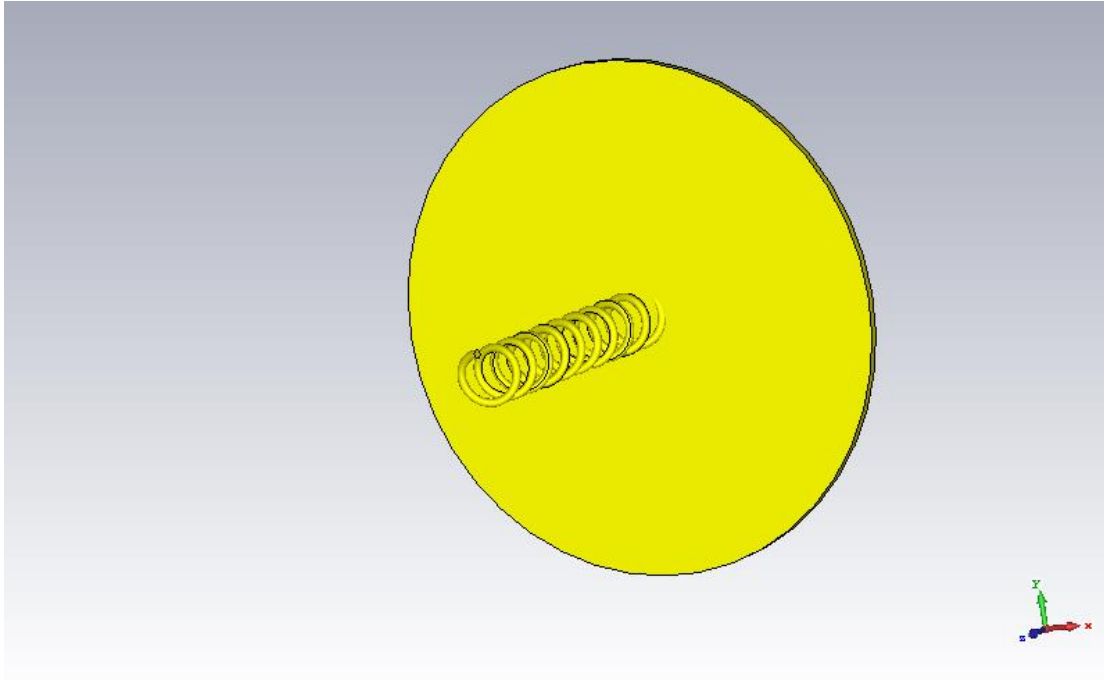
Waveguide Port



CST Design

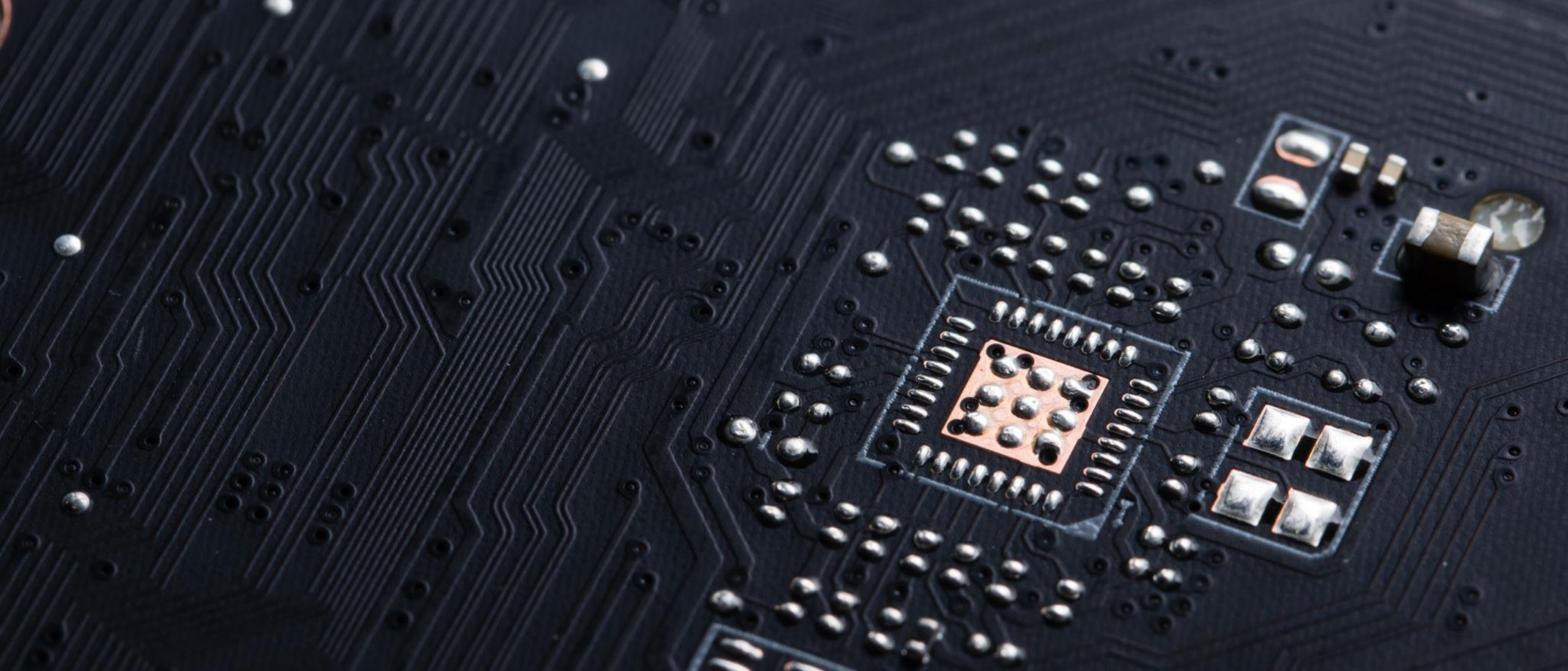


Antenna Designs



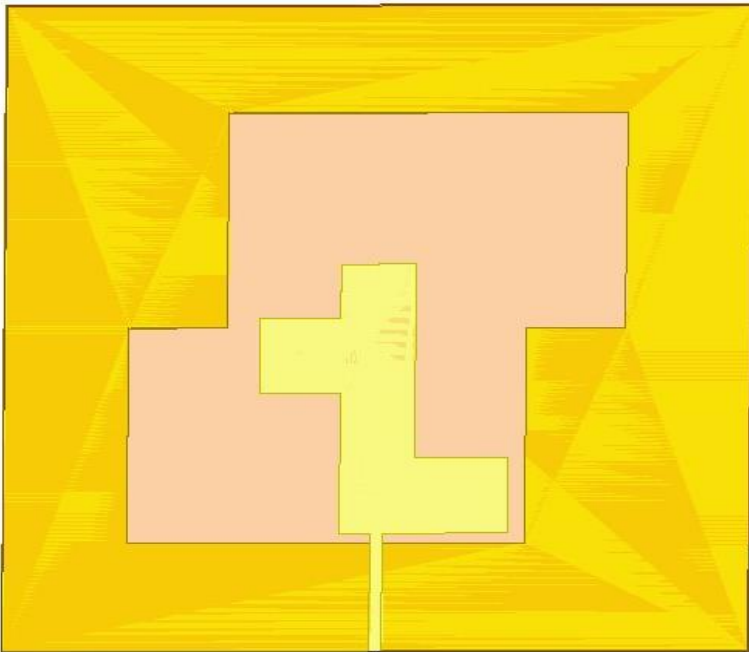
Phase 2: Core Design and Initial Validation (done)

- **Design Reproduction:** CST modeling of substrate, L shape Radiator, microstrip feed.
- **IBW Validation:** Verified $S_{11} < -10\text{dB}$ over 3-4.2 GHz ✓
- **CP Setup:** Far-field & axial ratio configuration
- **AR Validation:** Axial ratio simulation ($\leq 3\text{ dB ARBW}$)

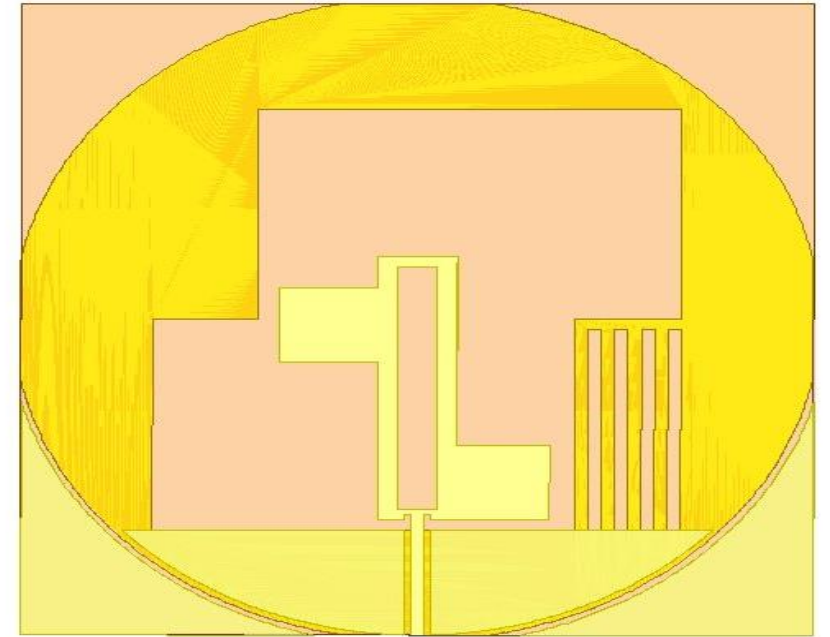


Coplanar Waveguide Antenna(CPW)

CST Design



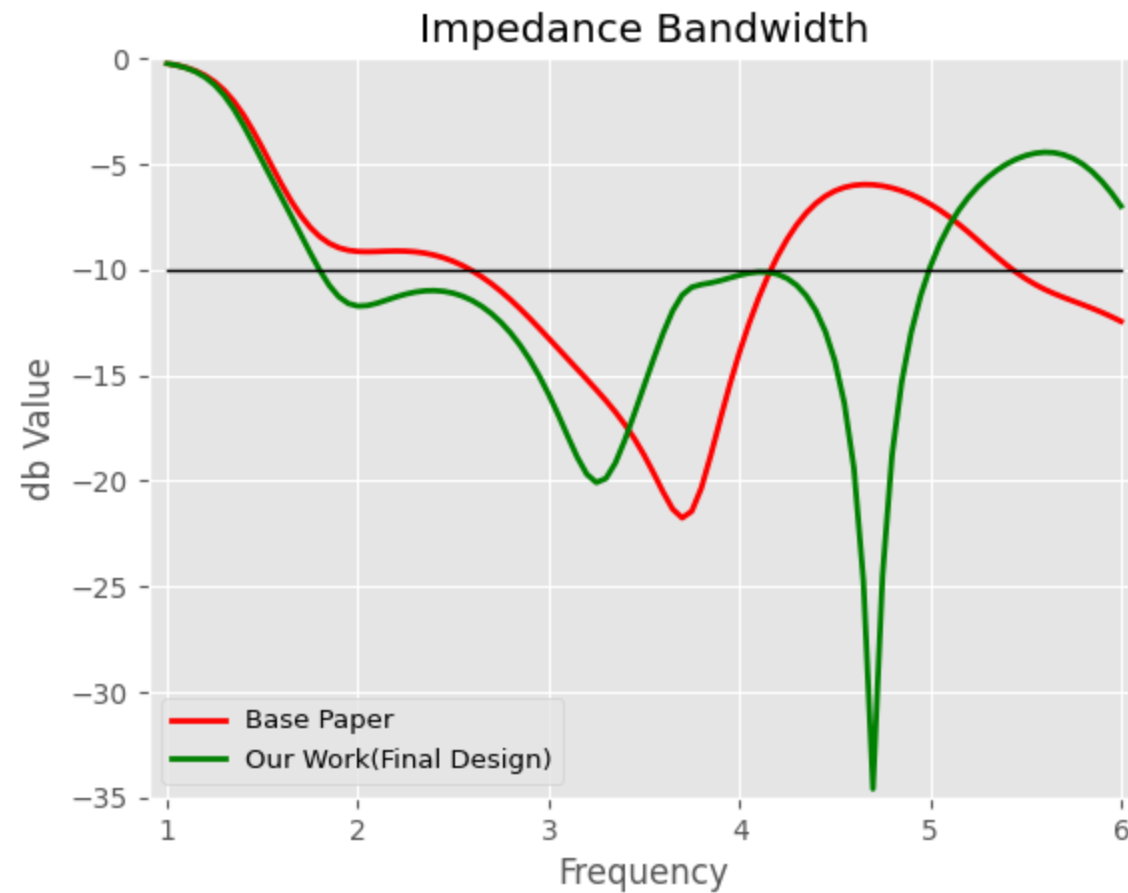
Base Paper Design

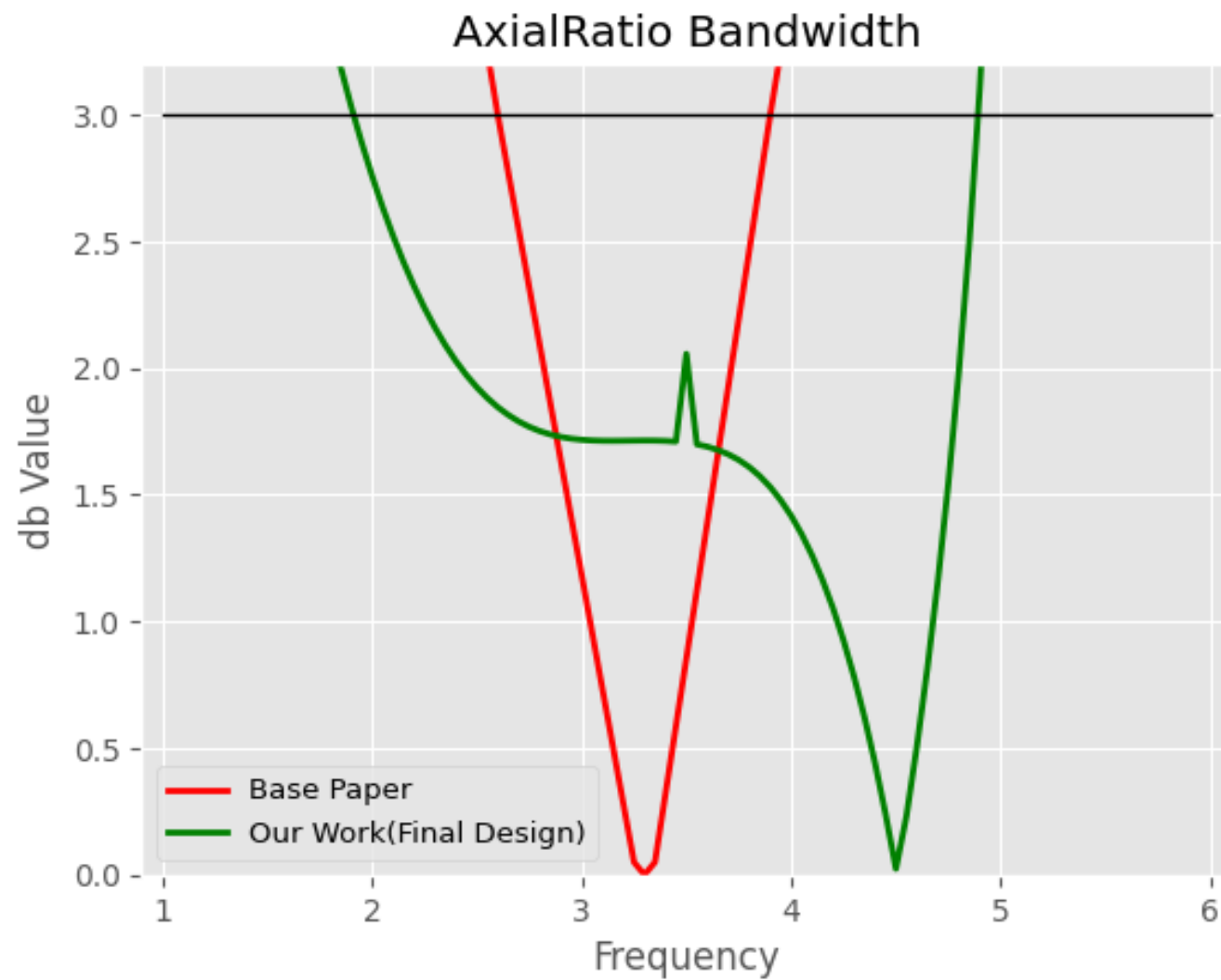


Our Design

Results

S11 Plot





Axial Ratio

Phase 3: Performance Optimization and Refinement (Done)

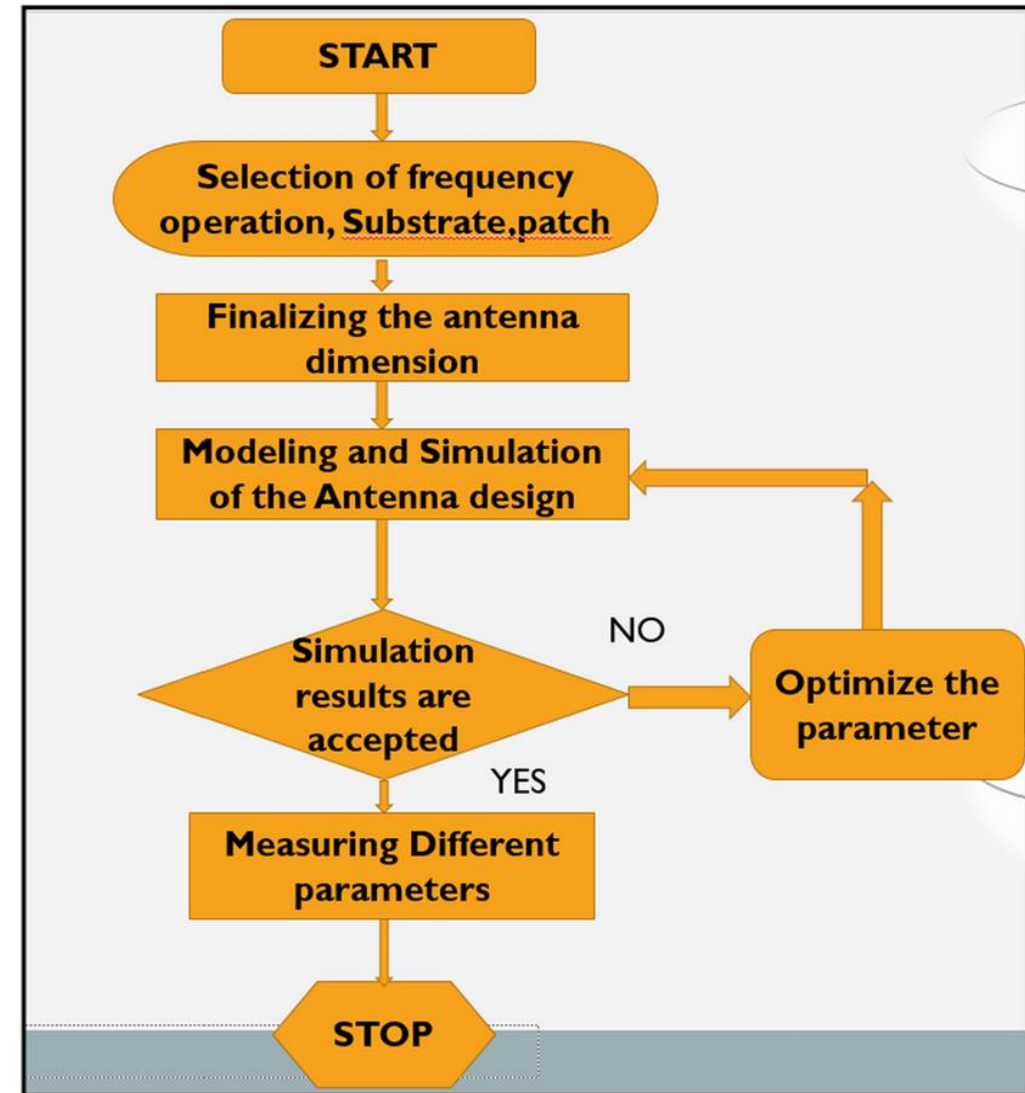


- **Parametric Analysis:** Study effect of key parameters on AR,, and IBW
- **Design Optimization:** UseHFSS optimizers to maximize ARBW within IBW
- **Port & Boundary Check:** Verify excitation and open-boundary conditions

Phase 4: Finalization and Reporting (Done)

- **Efficiency Analysis:** Check the useful Axial Ratio (Impedance & Axia Ratio Bandwidth Overlap)
- **Final Documentation:** Complete FYP report with results & optimization
- **Conclusion & Future Work:** Result comparison and future directions(e.g., fabricating a prototype, integration with RF circuitry, MIMO/Array Study,Parameter Optimization using Neural Networks)

FlowChart



Project Completion Timeline

Literature Review	September (2025)
How to use Software	October (2025)
Basic Designing	October (2025)
Research Paper Reproduction	November (2025)
Self designing & Optimizing	December (2025)
Fabrication & Testing	January-February (2026)
Thesis	March-April (2026)

Cost Analysis

Prototype fabrication and testing will be done at the Radar Seeking Department, AWC, through contacts of the Project Supervisor Dr Bilal Aslam.

Modern Tools & Software Requirements



HFSS/CST, Matlab, Origin Pro

Sustainable Development Goals (SDG)

- **Goal 9:** Build resilient infrastructure, promote sustainable industrialization and foster innovation
- **Goal 11:** Make cities and human settlements inclusive, safe, resilient and sustainable

Conclusion

- Developed a compact, wideband circularly polarized antenna for 28 GHz mm-Wave 5G/IoT applications.
- Planar, single-fed design fully modeled and validated through simulations.
- Achieved wide impedance bandwidth with $S_{11} < -10\text{dB}$.
- Ongoing optimization of axial ratio and realized gain for final CP performance.
- Final design targets a cost-effective solution with improved polarization matching and reduced multipath effects.

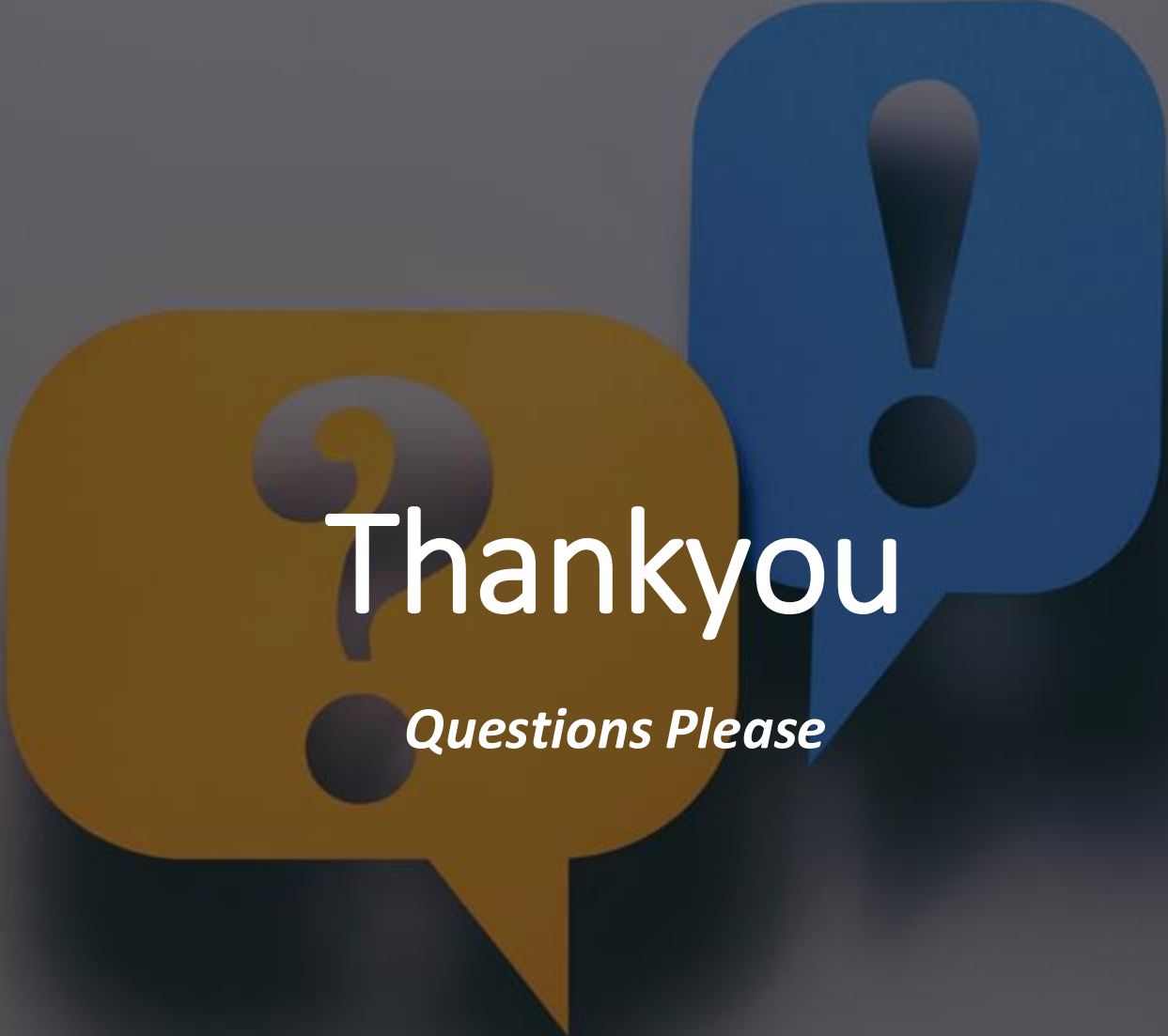
Core References

Primary Design Reference (The Base Paper):

- Ullah, U., Koziel, S., & Pietrenko-Dabrowska, A. (2022). "*Highly integrable planar-structured printed circularly polarized antenna for emerging wideband internet of things applications in the millimeter-wave band.*" *Scientific Reports*.

Foundational Antenna Theory:

- Balanis, C. A. (2016). *Antenna Theory: Analysis and Design*. Wiley-Interscience. (Essential for basic principles, gain, radiation patterns, and CP theory.)
- Lee, K. F., Luk, K. M., & Dahele, J. S. (2011). *Microstrip Patch Antennas*. Research Studies Press Ltd. (Necessary for understanding microstrip/CPW fundamentals and CP generation techniques.)



Thankyou

Questions Please